

Lecture 14: Seismic sources

David Al-Attar

Michaelmas term 2024

Outline and Motivation

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- We will then modify the theory to account for displacement across a fault.
- This is known as the **kinematic source approximation**, as the fault displacement is a specified parameter, not a dynamic variable.
- Finally, this framework allows us to introduce the concept of **inverse problems**: using seismic recordings to determine what happened at the source.

What Causes Earthquakes?

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- Most earthquakes occur near plate boundaries.
- The idea that earthquakes are produced by rapid motion across a fault was developed by H.F. Reid after the 1909 San Francisco earthquake.

Reid's Elastic Rebound Theory

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The Kinematic Approximation

Instead of modeling the complex physics of rupture, we will treat the motion on the fault as a known parameter and calculate the waves it generates.

A Problem With Our Theory

Our current theory for a hyperelastic body has a problem.

- The governing equations conserve the total energy of the body.

$$\frac{d}{dt} \int_M \left(\frac{1}{2} \rho ||\mathbf{v}||^2 + W(\mathbf{F}) \right) d^3x = 0$$

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Conclusion

According to our theory of hyperelasticity, a body at rest must remain at rest forever. Earthquakes are impossible!

Modifying the Theory for a Source

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To allow for earthquakes, we must modify the potential energy.

- We must break the **time-translation invariance** of the system, because an earthquake has a specific onset time.
- We do this by allowing the strain energy function W to have an explicit time dependence: $W(\mathbf{x}, t, \mathbf{F})$.
- We propose a specific form for this, separating the material's elastic properties (U) from the seismic source ($\bar{S}_{ij}$):

$$W(\mathbf{x}, t, \mathbf{F}) = U(\mathbf{x}, \mathbf{C}) - \frac{1}{2} \bar{S}_{ij}(\mathbf{x}, t) C_{ij}$$

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Definition: The Stress Glut

The **stress glut** ($\bar{\sigma}_{ij}$) represents the difference between the stress predicted by the material's elastic properties and the true stress in the body.

$$\bar{\sigma}_{ij} = \underbrace{2F_{ik} \frac{\partial U}{\partial C_{kj}}}_{\text{Elastic Stress}} - \underbrace{T_{ij}}_{\text{True Stress}}$$

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Physical Meaning

The stress glut can be interpreted as a failure of the elastic constitutive relation. This is consistent with the idea that material near a fault does not behave elastically during an earthquake.

Linearised Equations with a Source

Including the time-dependent source term in our linearised theory yields a new equation of motion.

Linearised Equation with a Source

The seismic source enters the equation as an effective body force:

$$\rho \frac{\partial^2 u_i}{\partial t^2} - \frac{\partial}{\partial x_j} \left(A_{ijkl} \frac{\partial u_k}{\partial x_l} \right) = - \frac{\partial \bar{S}_{ij}}{\partial x_j}$$

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Boundary Condition with a Source

It also acts as an effective surface traction on the boundary:

$$A_{ijkl} \frac{\partial u_k}{\partial x_l} \hat{n}_j = \bar{S}_{ij} \hat{n}_j$$

Note: In the linearised theory, we use $\bar{S}_{ij}$ to denote the stress glut.

Modeling an Idealised Fault

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- We model the fault as a mathematical surface, Σ , across which the displacement is discontinuous.
- This is a good approximation as long as the seismic wavelength is much larger than the width of the actual fault zone.
- We can write the displacement field $\mathbf{u}$ as a continuous part $\mathbf{u}^0$ plus a jump $\mathbf{w}$ across the fault plane (here, at $x_3 = 0$) using the Heaviside step function $H(x_3)$:

$$u_i(\mathbf{x}, t) = u_i^0(\mathbf{x}, t) + w_i(x_1, x_2, t)H(x_3)$$

Deriving the Stress Glut for a Fault

- When we differentiate the discontinuous displacement u_i to find the strain, the derivative of the Heaviside function produces a Dirac delta function $\delta(x_3)$.

$$\frac{\partial u_i}{\partial x_j} = \frac{\partial u_i^0}{\partial x_j} + \cdots + w_i \delta_{j3} \delta(x_3)$$

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- This delta function in the strain leads to a **singular**, non-physical term in the elastic stress.

The Stress Glut for a Fault

We define the stress glut to be precisely this non-physical term, so that it cancels out from the true stress. For a general fault surface Σ with normal vector $\mathbf{n}$, the result is:

$$\bar{S}_{ij} = A_{ijkl} w_k n_l \delta_\Sigma$$

Forward vs. Inverse Problems

The Forward Problem

Given a model of the Earth ($\rho, \mathbf{A}$) and a model of the earthquake source ($\mathbf{w}$), calculate the expected seismic waves (synthetic seismograms).

Model Parameters $\rightarrow$ Predicted Data

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The Inverse Problem

Given recorded seismic waves from seismometers, work backwards to determine the parameters of the model (e.g., what was the slip $\mathbf{w}$ on the fault?).

Observed Data $\rightarrow$ Model Parameters

Well-Posed vs. Ill-Posed Problems

There is a fundamental difference between forward and inverse problems.

Well-Posed Problems

A problem is considered **well-posed** if it satisfies three conditions:

- ➊ **Existence:** A solution exists for any valid input.
- ➋ **Uniqueness:** The solution is unique for a given input.
- ➌ **Continuity (Stability):** The solution depends continuously on the input (small changes in input lead to small changes in output).

A problem that fails any of these conditions is **ill-posed**.

Why Inverse Problems are Hard

- Any plausible physical theory, like our forward problem of wave propagation, must be well-posed. If it weren't stable, we could never test it experimentally due to measurement errors.

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Why Inverse Problems are Hard

- Any plausible physical theory, like our forward problem of wave propagation, must be well-posed. If it weren't stable, we could never test it experimentally due to measurement errors.
- In contrast, geophysical inverse problems are almost always **ill-posed**.
- The main reason is non-uniqueness. We are trying to determine continuous functions (like slip on a fault) from a finite number of measurements. The problem is fundamentally **underdetermined**.

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- A seismic source can be introduced into elasticity by modifying the strain energy, which gives rise to a **stress glut**.
- The stress glut for an idealised fault is proportional to the displacement discontinuity across the fault surface.
- A **forward problem** maps model parameters to data; an **inverse problem** maps observed data to model parameters.
- Inverse problems are typically **ill-posed** (non-unique), while forward problems representing physical laws must be **well-posed**.